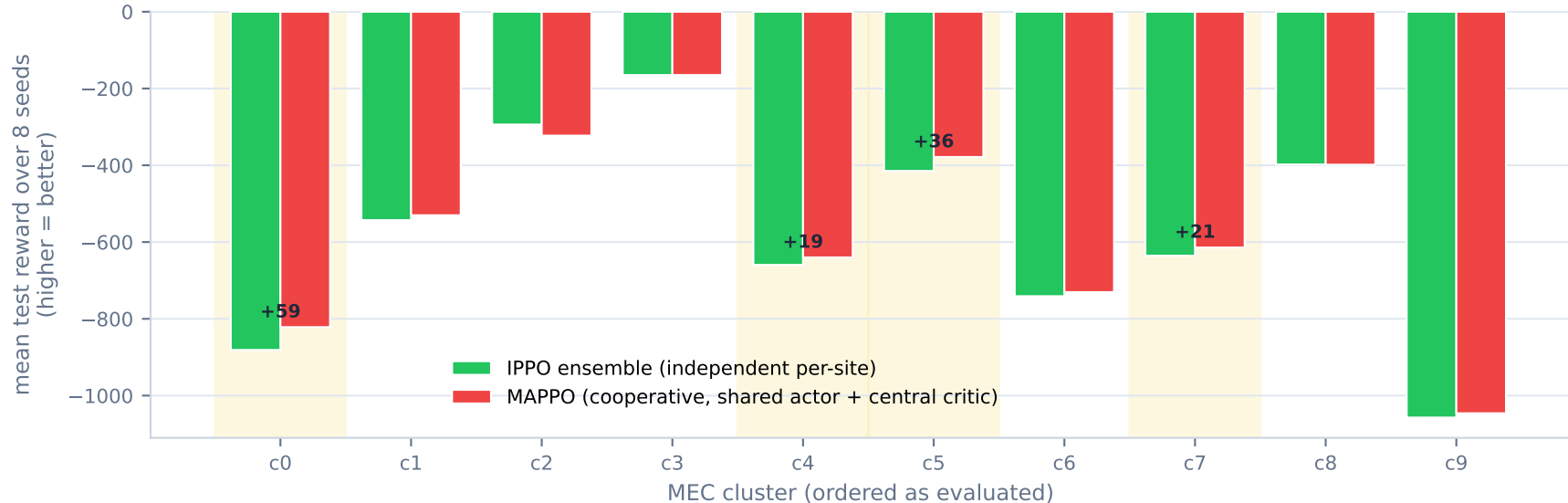


Per-cluster reward: independent single-site learning vs cooperative MAPPO



Shaded clusters: MAPPO > ensemble by >15 reward units — data-poor sites that inherit the operating rule from the fleet.